FORECASTING GOLD PRICE MOVEMENTS USING ALTERNATIVE DATA AND MACHINE LEARNING TECHNIQUES

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ABSTRACT

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Gold is an extremely important commodity as a store of value and a hedge against inflation. Its importance runs deep throughout history, to institutions as well as individuals alike. However, it is a notoriously volatile commodity, one that is easily impacted by geopolitical tensions and macroeconomic factors. Due to its importance, the ability to accurately predict its price movements would constitute immense value to governments, banks, and retail traders. This thesis explores the impact of incorporating alternative data such as google search trends, news sentiments and nontraditional financial metrics like the VIX and DXY indices into machine learning models for the purpose of forecasting short term gold price movements.

A probabilistic forecasting approach was adopted, focused on predicting price ranges rather than point values. Three models were used in this research: Prophet as a baseline linear time series models, and two tree-based machine learning models, Quantile Regression Forest (QQRF) and XGBoost with a quantile objective. A multi-horizon approach was used to forecast the price of gold over a ten trading day period. Four evaluation metrics were used to evaluate model performance. Continuous Ranked Probability Score (CRPS), Prediction Interval Coverage Probability (PICP), Average Interval width (AIW) and RMSE

The results show that the integration of alternative data led to an approximate 2% decline in predictive performance based on average CRPS scores. Upon deeper analysis into feature importance, it was discovered that traditional price based indicators largely dominated model performance, while alternative data features had a very limited and minimal impact on forecasting outcomes.

These findings indicate that alternative data, within the scope of this study, does not provide a statistically meaningful improvement in short term gold price forecasting. Despite the lack of forecasting improvement, the results highlight that not all forms of alternative data are equally useful for predicting gold prices, and that careful feature engineering and data selection remain critical.

CONTENTS

GLOSSARY

Concept	Definition
AIW	Average Interval Width
ARIMA	Autoregressive Integrated Moving Average
CRPS	Continuous Ranked Probability Score
DXY	US Dollar Index
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NLP	Natural Language Processing
PICP	Prediction Interval Coverage Probability
QRF	Quantile Regression Forest
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
S&P	Standard and Poor’s
USD	United States Dollar
VIX	Volatility Index

1 INTRODUCTION

Gold is a commodity of great historical importance. For thousands of years, it has been used as a medium of exchange, a store of value, and a hedge against inflation and other risks (Taneva-Angelova, Raychev and Ilieva, 2025). Although it has been a valuable commodity for thousands of years dating back to before ancient Egyptian times and despite its relatively high liquidity, it remains incredibly difficult to predict its price movements due to its high volatility and the many complex factors that drive its price behaviour. Experts typically tend to focus on geopolitical tensions, interest rates, inflation, macroeconomic factors, and currency exchange rate fluctuations to explain gold price movements (Taneva-Angelova, Raychev and Ilieva, 2025). However, the interactions between these factors are difficult to quantify and frequently lead to unpredictable short-term price movements. For that reason, understanding the dynamics and moving parts of gold price movements is extremely important for informed decision-making in international trade and investment strategies (John and Nidhina, 2025).

It is a well-known fact that gold price movements are influenced by many factors beyond simple supply and demand. Factors such as investor behaviour, collective sentiment, market expectations, media narratives and many others all have a significant, however indirect and subtle, effect on how gold moves, especially in the short term. In this context, public attention and market sentiment, exposed through search trends and news headlines, can act as early signals of evolving market conditions leading to a change in trajectory or an acceleration thereof when it comes to price movements. For example, a spike in search interest of terms relating to gold, safe-haven assets, or economic uncertainty may be a significant indicator of increased market fear, investor concern, and a potential bias towards safe-haven assets such as gold before these factors are actually reflected in gold's price action.

In recent years, advancements in big data and computational techniques have given rise to the availability of digital, behavioural, and unconventional economic indicators, otherwise known as alternative data. This presents an incredible opportunity to improve financial forecasting by integrating unconventional datasets, such as public sentiment, news articles, and search query trends, to uncover hidden patterns and correlations that traditional financial indicators might overlook (Cohen and Aiche, 2023). These data sources have been increasingly explored in the context of stock markets and cryptocurrencies, but there is comparatively limited research on applying them to commodity markets, especially gold.

Alternative data is often too large, too multidimensional, and too complex for traditional linear models to handle well. These models do not have the necessary complexity to take into account subtle nuances that may otherwise be significant. However, machine learning models are built to work with nonlinear patterns, variable interrelations, and added external features which allows them to be especially useful in detecting predictive signals. In recent years, machine learning methods have significantly advanced and become increasingly more accessible, and are now widely used in predictive modelling, pattern detection, and handling nonlinear relationships between datasets. When combined with alternative data, machine learning models might have the opportunity to uncover concealed correlations and interactions that traditional econometric models might have missed. This fusion of alternative data with sophisticated machine learning techniques offers a promising avenue for developing more robust and accurate gold price forecasting models (Sarvaiya and Ramchandani, 2022, p. 169).

This thesis places itself at the intersection of these developments by exploring whether alternative datasets, once processed through machine learning models, can contribute worthwhile improvements to short term gold price forecasts. To clarify, the purpose here is not to replace traditional financial and macroeconomic indicators or to claim an absolute method of forecasting but rather to offer more value in complementing existing information and predictive methods. In doing so, the research aims to contribute to the growing literature on data-driven forecasting and predictive analytics while maintaining a practical focus on interpretability and real-world relevance.

2 BACKGROUND

2.1 Importance of Predicting Gold Prices

Forecasting gold accurately is extremely important for various stakeholders, including retail investors, financial institutions, fund managers, as well as national economies, since gold serves both as a key investment asset and a vital industrial raw material (Trivedi, Somvanshi and J, 2022; Kumar, 2024). In addition to its investment and industrial capacity, gold's role as a safe-haven asset during periods of economic uncertainty highlights the importance of accurate forecasting for hedging against market volatility and preserving capital (Taneva-Angelova, Raychev and Ilieva, 2025). It is for these reasons that even marginal improvements in predictive accuracy can have major implications for decision-making, hedging, and risk exposure.

The issue however is that predicting fluctuations in the price of gold in the short term can be extremely challenging due to the fact that it reacts very quickly to changing market conditions. Unlike stocks and other financial instruments, gold does not produce any earnings or cash flow, therefore traditional valuation models would not apply. Its value is rather determined by opportunity cost, investors' risk appetite, and market expectations. Other factors such as central bank interest rates, inflation outlook, currency strength, especially that of the USD, as well as market volatility also have a significant, complex effect on the price of gold which makes forecasting the price of gold all the more difficult.

2.2 Current Limitations

Despite its significance, the field of gold price prediction still exhibits notable research gaps, particularly concerning the comprehensive integration of diverse influencing factors (Taneva-Angelova, Raychev and Ilieva, 2025). While extensive research has focused on traditional economic indicators, there remains a deficiency in studies that systematically incorporate alternative data sources and advanced machine learning techniques to capture the multifaceted dynamics influencing gold prices (John and Nidhina, 2025; Taneva-Angelova, Raychev and Ilieva, 2025). A lot of the existing research focuses mostly on historical price data and traditional technical indicators. Although these approaches have naturally had some success, it is our view that they fail to capture a holistic enough overview of the situation, leading to an oversight of potential signals stemming out of behavioural data. Furthermore, the research studies that do take into account alternative datasets, quite often incorporate them within the context of predicting movements in cryptocurrency or equity markets, while ignoring the world's oldest store of value, gold.

Another limitation in existing research is the tendency to prioritise forecasting accuracy without considering the value in marginal contributions. Small improvements in forecasting performance are sometimes ignored and judged as insignificant, despite their potential importance in financial decision-making contexts where even modest gains can translate into meaningful economic value. It is our view that even a small improvement in forecasting accuracy can have rippling effects on people's abilities to plan ahead, and make better financial decisions, at least in the short term. In a world where the future is always uncertain, one would greatly benefit from being even one percent more confident in what will happen in the future. This thesis addresses this issue specifically by exploring whether the integration of alternative data provides a meaningful impact on predictive performance, even when improvements are minor or incremental. This will be done by focusing strictly on short term forecasts, within a range of two to 4 weeks in the future, as well as comparing forecasting machine learning models with and without the inclusion of alternative datasets in an effort to better understand more clearly the practical value, if any exists, of these alternative datasets on predicting the price of gold. This background section outlines the rationale and motivation behind using a machine learning based approach on this research, as well as the need to provide empirical evidence of the role and value of incorporating alternative datasets into predicting commodity markets.

2.3 Aim of Thesis

The overall aim of the thesis is to explore whether alternative data that has been processed and analysed by machine learning models can enhance the prediction of short term (2 - 4 weeks) gold price movements.

This aim focuses on the predictive value of alternative data, not on building a deployable trading system.

2.4 Research Questions

Specifically, this research will address the following key questions:

- Which machine learning models perform best when incorporating alternative data for predicting gold price movements?
- Which types of alternative data, such as Google Trends data or sentiment analysis from financial news, have the strongest predictive relationship with gold price movements?
- Does the inclusion of alternative data improve forecasting accuracy, even when the improvement in predictive accuracy is small?

3 LITERATURE REVIEW

As previously mentioned, there have been many studies conducted that explain the importance of forecasting the price of gold for a multitude of reasons. Those same studies have explored various methodologies for achieving this purpose such as using ML models, deep learning models, classical forecasting methods, feature engineering, single and multifactored forecasting to name a few. The vast majority of which, however, have mostly focused on historical price data and traditional economic metrics as their data sources, very few have touched base on the integration of alternative datasets and their role in improving the precious metal's price forecasting such as Taneva-Angelova et al. (2025). This section will dive deeper into those studies, the data they used, their findings and accomplishments, the limitations they faced, and how they all synthesize and relate both to each other as well as to this research.

3.1 Early Research and Challenges

The concept of attempting to forecast the price of gold is not new. Researchers have been attempting to find the perfect recipe for accurate and sustainable gold price prediction for years considering the overwhelming benefits that it would entail. Early research has focused mostly on traditional econometric and statistical techniques such as linear regression and autoregressive models. These approaches have had some success, especially in periods of economic and geopolitical stability. However, they have often struggled to adapt to the nonlinear and volatile nature of gold price action during periods of volatility, structural breaks, geopolitical tension, and regime changes (Sarvaiya and Ramchandani, 2022). This limitation was a main driver of shifting towards a more adaptable and flexible forecasting approach capable of capturing the dynamic trends and complex variable interactions affecting gold price movements.

Despite this pivot, forecasting the price of gold accurately, especially in the short term, continued to be elusive. Masud et al. (2025) interestingly noted that gold's sensitivity to influencing factors such as inflation, market risk outlook, and interest rates, makes short term forecasts inherently difficult due to the rapidly changing nature of those factors. The challenge of finding the right balance between choosing which variables to consider while avoiding too much unnecessary noise continues to be a work in progress for many researchers as is evident by the continuous interest and research done around this topic. It is for this reason that this thesis believes that any incremental improvement in forecasting accuracy presents incredible value not only to the academic community but also within a practical context to investors and financial advisors alike.

3.2 Traditional Approaches to Gold Price Prediction

As previously mentioned, many early research studies have approached forecasting the price of gold using traditional econometric and classical forecasting models such as ARIMA and deep learning models like LSTM. ARIMA models, which stands for Autoregressive Integrated Moving Average, are powerful statistical models that are commonly used for analysing and forecasting time series based datasets. They achieve this by combining 3 main parts:

1. Autoregressive, which refers to past values. Auto, meaning self, regression analysis.
2. Integrated, which refers to the order of differencing applied to a non-stationary time series dataset to make it stationary. Stationary here means statistical properties, such as mean and variance, do not change over time. This is a core requirement of working with ARIMA models as they need to cancel out trends and seasonality in order to work properly.
3. Moving Average, representing the number of past forecast errors. To simplify, predictions by nature are never perfectly accurate. The errors made in previous predictions are captured by the moving average and used to enhance the accuracy of future predictions.

ARIMA models are a very common tool to linearly predict time series data, which means within the context of gold forecast, they perform somewhat well under stable conditions and periods of predictability. Their performance declines however in periods of high volatility or unexpected changes.

The second type of model is LSTM, or Long Short-Term Memory. LSTM is a kind of recurrent neural network (RNN) which is designed to learn long term dependencies in sequences of data such as text, speech, or time series. Essentially, it is able to remember important information for a long period of time while also forgetting what isn't important or useful.

Trivedi, Somvanshi and J (2022) were some of the researchers who have approached forecasting the price of gold using classical forecasting models such as ARIMA. They have found, unsurprisingly, that linear models only do somewhat well in forecasting gold price action in periods of stability and predictable growth. However, they fail to capture the nonlinear characteristics of a volatile, unstable market. Sarvaiya and Ramchandani (2022) took this research a step further by comparing ARIMA and LSTM models. Their findings were similar to that of Trivedi et al. in the limitations of ARIMA predictions of gold, but they were able to discover that LSTM is a far better model to use in gold price predictions. They used RMSE, or root mean square error, as an evaluation method and found that LSTM scored a value of 0.038 compared to ARIMA's 4.043. RMSE is a standard metric used to measure how far off predicted values are from actual values. In simpler terms, it is a "score" of how accurate a forecasting model is. The lower the score, the more accurate the predictions.

3.3 Machine Learning in Gold Price Predictions

The application of machine learning (ML) in time series forecasts, especially in predicting the price of gold, have increased drastically over the past few years. As mentioned in the previous paragraph, many studies find that ML models far outperform classical statistical methods when it comes to forecasting accuracy, especially when dealing with nonlinear, complex, timeseries. Cohen and Aiche (2023) explored multiple machine learning methodologies for gold price prediction and have concluded that models such as Random Forests and neural networks have a much better performance compared to linear models. They were able to discover that ML models are particularly suited to capturing hidden correlations between variables, especially when multiple market affecting factors are at play simultaneously. To clarify, we have briefly covered a subset of neural networks when we described RNNs such as LSTM. In a nutshell, neural networks are a type of machine learning technology that mimics the human brain. They use interconnected layers of nodes (neurons in the brain) to process and analyse data, recognize patterns and trends, and make predictions. Neural networks are the basis for deep learning models and have a wide array of applications such as visual recognition software, chatbots, and predictive analytics. Random Forests is another machine learning method used to make predictions by combining the results of many decision trees. It works by training each decision tree on a random subset of the data and flows through decisions until it makes a prediction. It then combines the results of all the trees, meaning the forest, by averaging the results for regression analysis or taking majority votes for classification to produce the final prediction. Hence the name, Random Forests.

Alharbi (2024) conducted a similar study where he utilized machine learning techniques to forecast the price of precious metals, specifically gold, silver, palladium, and platinum, from 2017 to 2023. He echoed prior findings that ML models are able to adapt to fast moving market dynamics which is further proof that machine learning can be employed not only as a predictive tool but also as a way of understanding underlying market correlations.

Despite these optimistic findings, many authors warn against treating advanced machine learning models as a “black box” solution. Although these models may have better performance and produce more accurate results, they must be evaluated and interpreted carefully especially in investment and financial market contexts where contexts and data reliability are crucial for trust and informed decision-making (Masud et al., 2025).

3.4 Alternative Data in Financial Forecasting

Beyond advancements in predictive analytics using machine learning and modelling techniques, many studies increasingly highlight the importance of considering alternative data in financial predictions. As previously mentioned, alternative data refers to unconventional datasets. In the context of this specific research, these include public sentiment, news articles, and search query trends that may shine a light on market expectations and investors' risk appetite (Cohen and Aiche, 2023).

Several studies such as that of Preis, Moat and Stanley (2013), have shown that sentiment data and search trends can act as strong influencing factors to market movements. These signals often come before noticeable changes in price, structure, or volatility in financial markets which makes them especially relevant for short term predictions. Taneva-Angelova, Raychev and Ilieva (2025) noted that alternative data factors can capture hidden market behaviour changes that traditional indicators often miss.

Despite this finding, research on incorporating alternative data factors into gold price prediction remains largely under researched. Divya Sri et al. (2024) came closest by integrating multiple factors, beyond historical gold prices, such as oil prices, the Standard and Poor's (S&P) 500 index, Dow Jones Index, US Bond rates (10 years), Euro to USD exchange rates, and prices of precious metals such as silver, platinum, palladium, and rhodium into their prediction model. However, the primary focus continued to be on conventional financial indicators. Studies that take into account behavioural or sentiment-based data as a primary input remain limited and few, which is a noticeable research gap that this thesis aims to address.

3.5 Sentiment Analysis

In order for news headlines and articles to be used for sentiment analysis, their qualitative information must first be quantified into numerical values that machine learning models would be able to process and understand, typically through the use of natural language processing (NLP) techniques.

NLP is a subset of artificial intelligence that allows computers to understand, generate, interpret and generally work with human language. NLP models have a wide variety of applications in daily life, from autocorrect and voice assistants like Siri on your phone to translators and chatbots such as ChatGPT. In this context, NLP models are able to process text in news headlines and articles, extract relevant information and determine their sentiment polarity (positive or negative), then assign a numerical sentiment score that can then be correlated to the price of an asset, such as gold. To clarify with an example, a news article headline talking about geopolitical instability in some part of the world would likely be assigned a negative sentiment score. This would infer increased market fear and could very well correlate to an increase in the price of gold since investors would flock towards safe haven assets. One would naturally expect that integrating such sentiment metrics into forecasting models, would enrich the prediction to allow a better understanding of market psychology and investor expectations, which would consequently capture the impact of such alternative data on price movements that traditional price-based models might miss.

This was the opinion of Masud et al. (2025) as they note that sentiment metrics can improve the accuracy of forecasts when combined with traditional time series techniques, especially during times of market instability. However, using sentiment data is not without its own challenges. Differences in data sources, preprocessing methods, and sentiment scoring techniques make study comparisons difficult. Furthermore, context matters, as many studies focus mainly on employing such alternative data to predict equity or cryptocurrency markets, leaving commodities such as gold under researched.

3.6 Model Evaluation

To ensure the reliability of machine learning models used in forecasting, it is crucial to have some evaluation metrics that would measure the performance of said models in terms of predictive accuracy. Metrics such as mean absolute error (MAE), mean absolute percentage error (MAPE), and RMSE are commonly used and provide a straightforward assessment of model performance by measuring the average magnitude of errors and their significance versus actual price values (John and K. Nidhina, 2025; Kumar, 2024; Trivedi, Somvanshi and J, 2022). RMSE was previously discussed, below is a brief explanation of MAE and MAPE.

Mean Absolute Error (MAE): this is a measure of how far off predictions are from actual values using the average of absolute differences. For example, if a predicted value is 8 and the actual value is 10, the absolute difference would be

2, represented by the formula: (abs(8-10)=2). This process would be done to all prediction points and then averaged to get the MAE.

Mean Absolute Percentage Error: following the same concept of MAE with the slight difference that the measure is in percentages. Absolute difference values are turned into percentages and then averaged.

Moreover, it is occasionally sufficient, and arguably more important, to measure directional predictive accuracy rather than absolute forecasted values. This is where directional performance metrics such Hit Rate or Sign Accuracy come in. These are metrics that are commonly used to determine whether a model correctly predicts the direction of price movement (up or down) which can be of great importance to traders specifically.

While all these metrics are crucial for model comparison and assessment, they overlook the importance of small forecast improvement, which is a recurrent issue across the literature. Trivedi et al. (2025) noted that marginal gains, or small improvements, in predictive performance can have an important effect on decision making, especially in financial contexts where decisions are often made with uncertainty and have significant impact. This perspective is more relevant in short term gold price forecasting. Given market volatility and sensitivity to world events, expecting perfect or large improvements in predictive accuracy would be unrealistic, whereas exploring whether alternative data provides incremental improvements may be a lot more appropriate as a research goal.

3.7 Research Gaps

The compiled literature revealed many interesting insights as well as some significant research gaps. This section will briefly outline the relevant research gaps that have been identified.

1. Despite the comprehensive research done on employing machine learning techniques in forecasting financial markets, a clear gap was identified in incorporating alternative datasets into forecasting the price of commodities, particularly gold.

2. When considering alternative data sources for forecasting, sentiment data was often overlooked. This is a clear gap given gold’s sensitivity to world news and investors’ risk tolerance.
3. Most research focus on extremely narrow forecasting horizons such as next day predictions, whereas none have considered a broader short-term forecast covering multiple weeks.
4. There is limited consideration of the importance of marginal improvements in forecasting accuracy.

4 DATA DESCRIPTION

For the purpose of conducting this research, certain datasets needed to be attained. This section will discuss each of the five datasets used, how they were collected, how they were pre-processed and cleaned, as well as a data dictionary describing each dataset's fields.

4.1 Data Collection

There was a total of five publicly available, free, datasets that were collected for the purpose of this research. All of these datasets were collected for the time period starting on the first of January 2008, and ending on the thirty first of December 2017, covering a total period of ten years.

4.1.1 Gold Historical Data

The first data set is historical values of the price of gold. This dataset was manually downloaded in the form of a csv file from investing.com, a leading financial data platform founded in 2007. The dataset contains daily historical XAU/USD opening, closing, daily high, and daily low prices from the period of 2008-01-01 to 2017-12-31. This dataset contains 2607 rows and 7 columns. Table 1 shows a data dictionary describing the dataset's different fields. This table shows the dataset's original fields and the number of records corresponds to the number of trading days in the above mentioned 10 year period. It is worth noting that not all fields are relevant for the purposes of this research, specifically the volume and percent change fields.

Table 1. Data dictionary of historical gold values dataset.

Field	**Data Type**	**Description**
Date	datetime	The date of the observed values

Price	float64	The closing price for that date
Open	float64	The opening price for that date
High	float64	The highest price of gold on that date
Low	float64	The lowest price of gold on that date
Vol.	float64	The volume traded in USD
Change %	float64	The percentage change in price from previous day closing

4.1.2 Google Trends Data

Google trends and keyword interest over time was accessed and collected through a python library called “pytrends”. The values extracted represent the popularity of each term on any given day. This popularity value is normalized on a scale of 0 to 100, 100 being peak popularity and 0 being insufficient data.

Pytrends is an unofficial Python library that automates the downloading and extraction of search trend data from Google Trends. It is widely used by data scientists, researchers, marketers, and analysts to access, analyse, and visualize public search data programmatically.

The list of keywords that are observed includes ‘gold price’, ‘gold forecast’, ‘buy gold’, ‘inflation’, ‘safe haven’.

Since Pytrends is an unofficial python library, its use comes with certain limitations. Specifically, sending a request to the google API with too many keywords will result in an error. Simultaneously, if we attempt to extract daily data for too large of a time period, Google's API will only return the data for the first of every month, which is not the granularity needed here. A potential work around is to use a loop to fetch the data for each keyword individually and break apart the time period over many requests to avoid time granularity degradation. This approach is demonstrated in appendix 1. The output of this data fetching is a popularity score for each term that is a measure of how popular this term was during that period, relative to its peak. For example, a score of 100 means peak popularity, signifying the most searches for this term during this period. A score of 50 means on that day, that term was half as popular compared to its peak, and 0 meaning insufficient data. This dataset contains 3654 rows and 6 columns. The number of records in this dataset corresponds to the number of days in the chosen 10 year period. Table 2 below shows a data dictionary describing the dataset's different fields.

Table 2. Data dictionary of google trends dataset.

Field	**Data Type**	**Description**
date	datetime	The date of the observed values
gold price	Int64	Popularity score for that term on that day from 0 - 100
gold forecast	Int64	Popularity score for that term on that day from 0 - 100
buy gold	Int64	Popularity score for that term on that day from 0 - 100
inflation	Int64	Popularity score for that term on that day from 0 - 100
safe haven	Int64	Popularity score for that term on that day from 0 - 100

4.1.3 News Sentiment Data

This is a news dataset for the commodity market, specifically gold, where over 10,000+ news headlines across multiple dimensions were manually annotated into various classes. The original dataset has been sampled from a period of 20+ years (2000-2021). The news sentiment data was downloaded from Kaggle.com

(Sinha and Khandait, 2021), which is the world's largest online community and platform for data science and machine learning, owned by Google.

According to the original authors, the dataset has been collected from various news sources and annotated by three human annotators who were subject matter experts. Each news headline was evaluated on various dimensions, for instance - if a headline is a price related news, then what is the direction of price movements it is talking about; whether the news headline is talking about the past or future; whether the news item is talking about asset comparison; etc. This dataset contains 10570 rows and 10 columns; each record corresponds to one news headline on any given day. Table 3 below shows a data dictionary describing the dataset's different fields.

Table 3. Data dictionary of news sentiment dataset.

Field	**Data Type**	**Description**
Dates	datetime	The date of the news headline
URL	object	URL of the news headline
News	object	News headline
Price Direction Up	int64	Does the news headline imply price direction up? Integer between 0 meaning no, and 1 meaning yes.
Price Direction Constant	int64	Does the news headline imply price direction sideways (no change)? Integer between 0 meaning no, and 1 meaning yes.
Price Direction Down	int64	Does the news headline imply price direction down? Integer between 0 meaning no, and 1 meaning yes.

Asset Comparison	int64	Are assets being compared? Integer between 0 meaning no, and 1 meaning yes.
Past Information	int64	Is the news headline talking about past events? Integer between 0 meaning no, and 1 meaning yes.
Future Information	int64	Is the news headline talking about future? Integer between 0 meaning no, and 1 meaning yes.
Price Sentiment	object	Price sentiment of gold commodity based on headline (positive, negative, neutral)

4.1.4 Volatility Index (VIX) Data

The volatility Index, also known as the VIX index, measures the market's expectation of future volatility, representing the expected speed and magnitude of price changes over the next 30 days. It is a commonly used index to gauge market fear, when investors' fear and anxiety increase due to market downturns or geopolitical instability, the VIX increases. This dataset was used as the market fear factor of the data collected and was downloaded from Yahoo Finance in the form of CSV. This dataset contains 2518 rows and 7 columns. Each record corresponds to an active trading day on the US market, which is where the VIX values are based on. Table 4 below shows a data dictionary describing the dataset's different fields.

Table 4. Data dictionary of historical gold values dataset.

Field	**Data Type**	**Description**
Date	datetime	The date of the observed values
Price	float64	The closing value for that date
Open	float64	The opening value for that date

High	float64	The highest value of VIX index on that date
Low	float64	The lowest value of VIX index on that date
Vol.	float64	The volume traded
Change %	object	The percentage change in value from previous day closing

4.1.5 USD Index (DXY)

The final dataset consists of historical values of the United States Dollar Index (DXY) for the stated time period. The DXY is a measure of the value or strength of the USD against a basket of six other international currencies. Specifically, these currencies are the euro, Japanese yen, British pound sterling, Canadian dollar, Swedish krona, and Swiss franc. This index is relevant due to its inverse relationship to the price of gold. When the value or strength of the USD decreases, gold becomes cheaper for foreign investors since its value is usually measured in USD, this typically leads to an increase in gold demand, driving the price of gold up. This dataset contains 2590 rows and 7 columns. Each record corresponds to an active trading day on the foreign exchange markets. Table 5 below shows a data dictionary describing the dataset's different fields.

Table 5. Data dictionary of historical gold values dataset.

Field	**Data Type**	**Description**
Date	datetime	The date of the observed values
Price	float64	The closing value for that date
Open	float64	The opening value for that date
High	float64	The highest value of DXY index on that date

Low	float64	The lowest value of DXY index on that date
Vol.	object	The volume traded
Change %	object	The percentage change in value from previous day closing

4.2 Data Processing

In order for the above-mentioned datasets to be readily processable for our purposes, they first need to go through certain preprocessing stages. Namely, the datasets need to be cleaned, data quality issues need to be addressed, and feature engineering needs to take place. These steps will be discussed in the following sections.

4.2.1 Data Cleaning

The primary data quality issue that needed to be addressed was with the news sentiment dataset obtained from Kaggle. An issue was discovered with the values in the “Dates” column, wherein date strings had inconsistent, and sometimes erroneous formats. For example, “0200-04-23” instead of “2007-04-23”. To address this critical issue, a python script was developed (Appendix 2). This script looped over the data and reformatted the values in the “dates” column into YYYY-MM-DD format as well as fixed data quality issues like the one mentioned above by scanning the news headline’s URL for a date and replacing the faulty value in the “dates” column. This approach fixed the vast majority of the data quality cases; the few remaining edge cases were fixed manually.

Furthermore, the following measures were also taken as part of the data cleaning phase:

- All date columns were converted into datetime data type and sorted in ascending values.

- All commas and percentage signs were removed from numeric columns.
- Numeric columns were converted to float data type.
- Sentiment data was filtered to the relevant period (2008-2017).
- Datasets were merged together using the date columns.

4.2.2 Data Transformation

After cleaning the data, it was transformed to prepare it for usage by the ML models. For this purpose, the following measures were taken:

- News sentiment data was aggregated to daily level.
- Daily net sentiment score was calculated
- Created various rolling features such as moving averages and standard deviations, which will act as inputs to the ML models
- Created lagged features to prevent data leaks. Data leakage is when future data that should not yet be available is used to train an ML model that will predict future values.
- Created multi-horizon targets for the next 10 trading days
- Produced two datasets, one with and one without alternative data such as google trends and news sentiment data.

This branching of source data into two separate datasets allows for a clear comparison between different models as well as highlighting the incremental value, if any, that alternative data potentially provides when predicting gold price movements.

5 METHODOLOGY

Once the data has been thoroughly cleaned and pre-processed, the next stage would be to utilize it to attempt to answer the previously stated research questions. The figure below visualizes the flow of data at different stages throughout this research in a simplified manner.

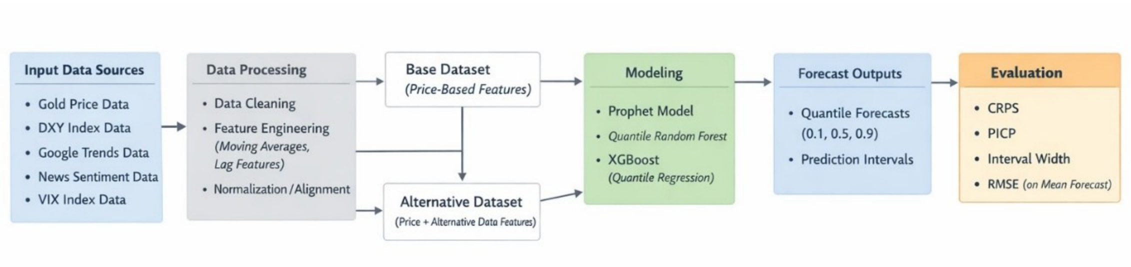

Figure 1. Gold forecasting data flow

This section will discuss in detail the chosen strategies and methodologies, describe the machine learning models as well as the evaluation criteria applied to the pre-processed data for gold price forecasting.

5.1 Forecasting Framework

This study uses a supervised machine learning framework to predict gold price movements over short term time horizons, ranging from 1 to 10 trading days in the future.

Supervised machine learning is a common machine learning training methodology used to train models. In simple terms, it is the equivalent of teaching a computer using examples, where models are trained using labelled data, meaning each input (feature) is mapped to an output (target). The machine learning model learns the relationship between these inputs and outputs and uses that knowledge to predict future values.

Instead of focusing primarily on forecasting point predictions, meaning specific price levels, this research adopted the probabilistic forecasting approach, with an emphasis on predicting price ranges. This approach allows for a more realistic evaluation of forecast accuracy as well as a more concrete assessment of future price trajectories by taking into account the inherent uncertainties, which is particularly relevant in volatile financial markets.

Furthermore, since the focus is on predicting short-term price movements in the next 1 to 10 trading days, it was decided to employ a direct multi-horizon approach. A direct multi-horizon approach is a forecasting method where a separate machine learning model is trained for each relevant future time step. In practical terms, this study will train 10 models to forecast each of the future 10 trading days, each model would learn the relationship between current day's features and its assigned horizon target. This approach was preferred for a couple of reasons:

1. Unlike many of the approaches mentioned in the literature review studies, this study will be focusing on a slightly longer time horizon, 10 trading days, as opposed to just one day in the future. Despite this still being classified as short term forecasting, the longer horizon increases the risk of imprecision. Training a separate model for each separate time horizon reduces that risk, resulting in a more accurate future prediction.
2. The alternative would have been rolling forecasts, wherein one model predicts the next future value, and then rolling it forward to predict the next future value after that. This approach would result in accumulated mistakes and compounding errors, which this research would ideally like to avoid.

As previously mentioned, two feature configurations are evaluated:

- Base model, which includes traditional financial indicators such as price-based features, VIX, USD index
- Alternative model, which includes all the base features with the addition of Google Trends and sentiment data variables

This setup enables controlled comparison of the potential incremental predictive value of alternative data. However, in order to preserve the temporal structure of the data and the integrity of this study, a chronological 80-20 split with lagged features was implemented to prevent look ahead bias and data leakage.

5.2 Machine Learning Models

To evaluate predictive performance, the following models are implemented:

- Prophet (baseline model)
- Quantile Regression Forest (QRF)
- XGBoost with quantile objective

This section will briefly describe each model and why it was chosen.

5.2.1 Prophet

Prophet is an open source, additive regression forecasting model developed by Meta, formerly Facebook (Taylor and Letham, 2018). This model was developed specifically to forecast time series in a simple and intuitive way. It works by breaking down a time series dataset into three parts:

- Trend, meaning the overall long-term direction.
- Seasonality, meaning repeating daily, weekly, or yearly patterns.
- Events, meaning special dates that could have an effect on the forecast such as holidays or economic or political announcements.

It then uses all three parts to provide a forecast.

This model was chosen for multiple reasons, the first of which is its ease of use. It does not require deep statistical or technical expertise and has a sort of "plug and play" architecture. The second reason is its ability to handle non-linear trends, seasonalities, and global events well, making especially suitable for forecasting financial markets or commodities such as gold.

This model will be used as the benchmark against which other models will be evaluated.

5.2.2 Quantile Regression Forest (QRF)

Due to this study's focus on probabilistic forecasting, choosing a quantile regression forest model was a natural inevitability. Simply stated, QRFs are a subset of random forest models that predict a range of possible outcomes rather than a single average value. It works by estimating any chosen quantile, in this case an upper and lower bound which serve as the prediction intervals, or conditional quantiles (Meinshausen, 2006).

This model was chosen for its ability to model non-linear relationship and interactions between multiple variables, making it ideal for the purposes of this research.

5.2.3 XGBoost

XGBoost, otherwise known as Extreme Gradient Boosting, is another tree-based powerful, open source, machine learning algorithm designed to handle structured, tabular data (Chen and Guestrin, 2016).

It is commonly used for both classification and regression predictions and preferred by experts due to its graceful handling of missing values, its efficiency and speed, and the fact that it provides feature importance scores, allowing researchers to see which inputs have the most impact.

It works by building a collection of decision trees, each acting as a mini model that cooperates with its peers. Each new tree attempts to fix the mistakes of its predecessors, this is the "boosting" part. The "gradient" part refers to the use of trial and error to optimize predictions step by step, making the model more accurate with each iteration. Finally, it uses regularization, or penalties, to prevent overfitting, thus ensuring the model doesn't just memorize the training data.

Unliked QRF, XGBoost is normally designed for point predictions, however in this case, its objective function is changed to quantile loss in order to predict quantile intervals. The quantiles used for both XGBoost and QRF will be 10th and 90th percentile resulting in an 80% prediction interval.

5.3 Evaluation Methods

As previously mentioned, this research is implementing a probabilistic forecasting approach in order to predict the future price of gold while incorporating alternative data features. To re-iterate and keep the research questions in focus, the objectives here are:

- Determine whether alternative data can improve accuracy.
- Determine which machine learning model performs best at predicting gold price movements.
- Determine which alternative data has the strongest predictive value.
- Determine whether the inclusion of alternative data, provides at least incremental forecasting improvements.

In order to achieve these objectives and answer this study's research questions, certain metrics must be measured to evaluate predictive performance. These metrics must measure the following:

- Coverage: whether actual prices fall within the predicted intervals ranges
- Sharpness: whether predicted ranges are sufficiently tight to be useful
- Accuracy: how close the predicted cumulative distribution is to the actual values

Based on these criteria, the following metrics have been chosen:

- Continuous Ranked Probability Score (CRPS) as a primary metric of accuracy
- Prediction Interval Coverage Probability (PICP) to measure coverage
- Average Interval Width to measure sharpness
- Root Mean Squared Error (RMSE) as a secondary metric

Forecast performance will be evaluated separately for each of the 10-day horizons using a walk forward method wherein models are trained on past data and tested against unseen future values, if a walk forward approach is not a viable option, a simple chronological 80-20 split will be used. The following sections will describe each metric in more detail.

5.3.1 Continuous Ranked Probability Score (CRPS)

CRPS is a measure of how good a probabilistic forecast is. It compares the predicted distribution to the actual observed value and produces a single score (Gneiting and Raftery, 2007). The lower the CRPS score is, the better the forecasting accuracy of the model. It is worth noting that this method rewards accuracy, calibrated uncertainty and appropriate spreads. Simultaneously, it penalizes overconfidence, which is when the intervals are too narrow, as well as excessive uncertainty, which is when the intervals are too wide. In this case, CRPS is approximated using discrete quantiles due to model constraints.

To put this in practical terms, for example, if a probabilistic weather forecasting model produces a forecast which indicates a 70% change of rain and it does rain, this will result in a lower CRPS score, which is good. Whereas if the same model forecasts a 50% chance of rain, the score will be higher, or worse, due to the increased uncertainty and lack of confidence.

5.3.2 Prediction Interval Coverage Probability (PICP)

PICP is a common measure of coverage in probabilistic forecasting. It measures how often actual values fall within predicted intervals (Khosravi, Nahavandi and Creighton, 2011). For example, suppose a model is forecasting 100 days of rain with 80% prediction intervals, if the forecast was true for 85 days out of the 100, the PICP is then 85%. Ideally the PICP should be as close as possible to the chosen prediction interval. For this experiment a PICP of 80% will be used.

It is important to note here that measuring coverage alone is not enough. Models can achieve 100% coverage simply by using unreasonably wide intervals such as “the price of gold tomorrow will be between $100 and $10000”, the probability of such a statement being accurate is almost always 100%. However, this sort of prediction has no predictive or practical value, which brings us to the next metric, average interval width.

5.3.3 Average Interval Width

As one can infer from the name, average interval width (AIW) is a measure of how wide the prediction intervals are. This is a measure of sharpness that refers to the average width of the predicted interval, or the distance between the upper bound and the lower bound (Khosravi, Nahavandi and Creighton, 2011). For example, if a model forecasts that the price of gold tomorrow will be between \$1900 and \$2100, the interval width would be \$200. The average interval width metric can be calculated using the following formula:

$$Interval\ Width = \frac{1}{N}\sum_{t=1}^{N}(Upper_t - Lower_t)$$

The lower the interval width, the more precise and confident a model is, however precision comes with the risk of missing coverage, meaning the actual value does not fall within predicted intervals. Therefore, is it important to balance between coverage and sharpness when designing a probability forecasting model.

6 ANALYSIS AND RESULTS

Following the guidelines outlines in the methodology section, a python notebook "thesis_modeling.ipynb" was created. Within this notebook, the thesis experiment was conducted with the following criteria:

- 3 models were used: Prophet, Quantile Regression Forest, and XGBoost Quantile.
- 80-20 chronological train-test split, with an 80% prediction interval.
- RMSE, CRPS, PICP, and AIW were used as evaluation metrics over a 10 day horizon.

The results of this experiment will be discussed for each model.

6.1 Prophet Model

As previously mentioned, prophet is a linear time series forecasting model, as such, it does not take into account any engineered features. The inputs are simply dates and values. The outputs would be a lower bound forecast, an upper bound forecast, and a mean point forecast. Naturally, since features are inherently not considered this means that there would be no difference in forecasted values between the base dataset and the alternative dataset. The following chart shows the model's forecast vs actual gold prices in the last 10 days of both datasets.

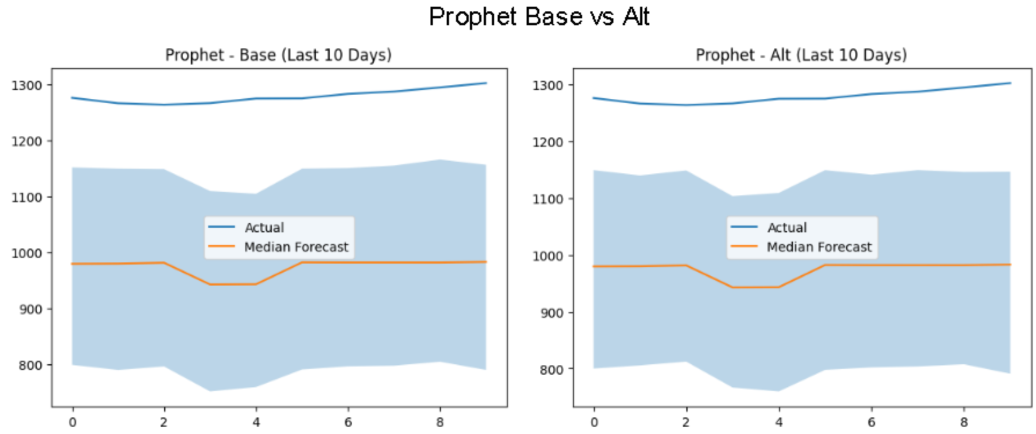

Figure 2. Prophet forecast base vs alternative datasets, last 10 days

In the above charts, The X-axis represents trading days while the Y-axis represents gold prices. The blue line represents the actual values of gold price during the 10 trading day period between the 20th of November 2017 and the 15th of December 2017. The actual values fluctuated between $1240 to $1280 per gold ounce. The lightly blue shaded area represents the upper and lower bound of the model's forecast, while the orange line represents the median or 50th percentile forecast.

The model had a wide average interval width of around $192 across the whole dataset, and an even larger ~$350 across the subset of the last 10 days shown in the figure above.

This wide gap between actual prices and forecasted range shown is to be expected considering 1. This model works best with linear values, while gold price movements are notoriously volatile and 2. Forecasts are known to be less accurate as the time horizon expands, the figure above shows the forecast of the last 10 days over a 10 year period. To further validate this reasoning Figure 3 below shows the forecast of the first 10 days which shows accurate forecasts for the begning of the forecasting horizon.

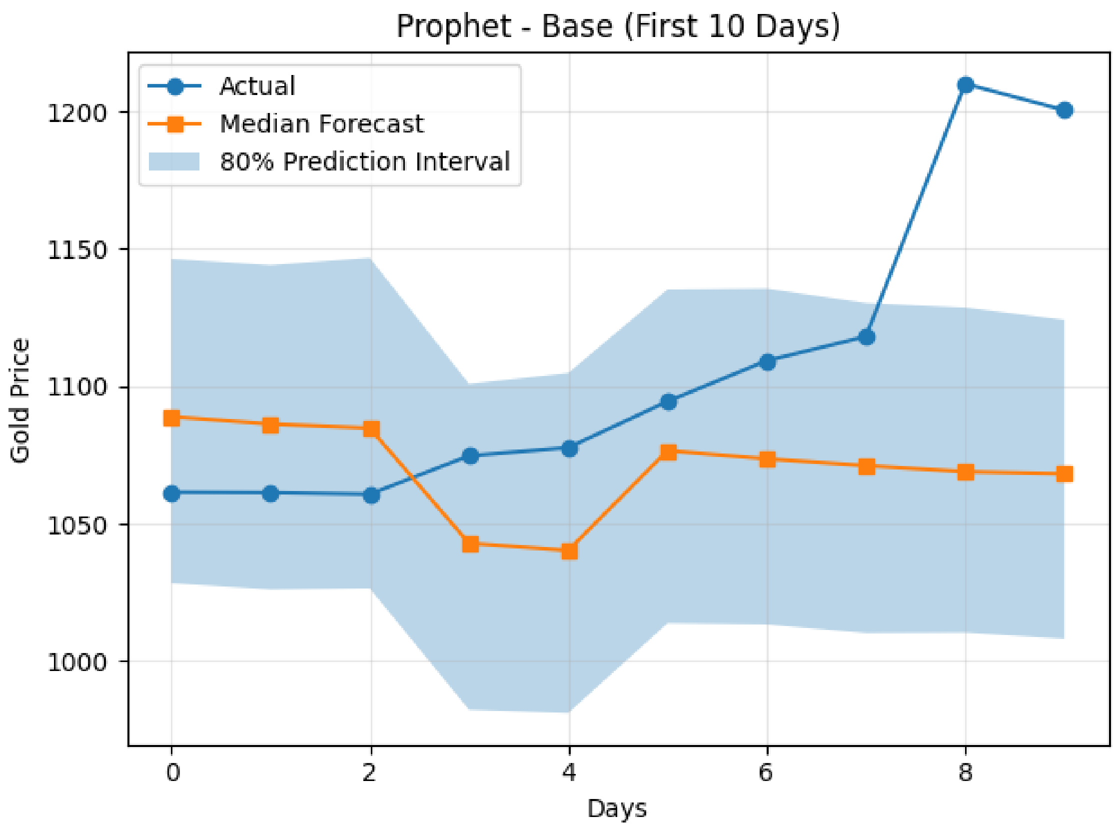

Figure 3. Prophet forecast, first 10 days

Beyond the average interval width, other metrics were measured such as the PICP measuring coverage, the CRPS measuring overall accuracy, and the RMSE measuring how far off the forecast is from actual values. As a contextual reminder, for the CRPS and RMSE metrics, the lower the score, the better the performance. Conversely, a higher PICP score is better as it means actual values fell within the forecasted range more often. The following figure showcases two tables showing all the metrics for the Prophet model across both base and alternative datasets for each of the 10 horizons on average.

Prophet - Base Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	90.838636	0.022663	194.026105	244.921109
2	93.824255	0.022663	192.034095	245.325327
3	93.035927	0.022663	193.822720	245.823539
4	92.386873	0.022663	190.831291	246.306164
5	92.698452	0.022663	194.190999	246.869683
6	93.795928	0.022663	189.798948	247.218381
7	93.550271	0.022663	194.629222	247.653422
8	93.331069	0.022663	194.528546	248.050532
9	93.220437	0.022663	200.059321	248.346693
10	94.042123	0.022663	193.141491	248.434416
Average	93.072397	0.022663	193.706274	246.894927

Prophet - Alternative Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	93.531998	0.025496	192.887419	244.921109
2	93.237994	0.022663	196.890252	245.325327
3	92.461944	0.022663	195.730123	245.823539
4	91.829691	0.022663	190.553390	246.306164
5	92.555029	0.022663	190.498178	246.869683
6	93.445410	0.022663	191.748030	247.218381
7	93.159889	0.022663	193.802142	247.653422
8	93.433121	0.022663	192.604033	248.050532
9	95.679466	0.022663	193.317661	248.346693
10	94.300566	0.022663	192.831631	248.434416
Average	93.363511	0.022946	193.086286	246.894927

Figure 4. Prophet model avg evaluation metrics per daily horizon of the test data subset (last 20% of 2007-2017 dataset). Base vs Alternative datasets

We can observe from the above figure that on average, the PICP, measuring coverage, only has a value of around 0.023. This means that on average, actual values fell within the forecast range only 2.3% of the time. Consequently, the RMSE shows a very high value of around 247 points while the CRPS also has a high score of approximately 93 points. All of these metrics can be visualized in a bar graph for comparison in figure 5 below.

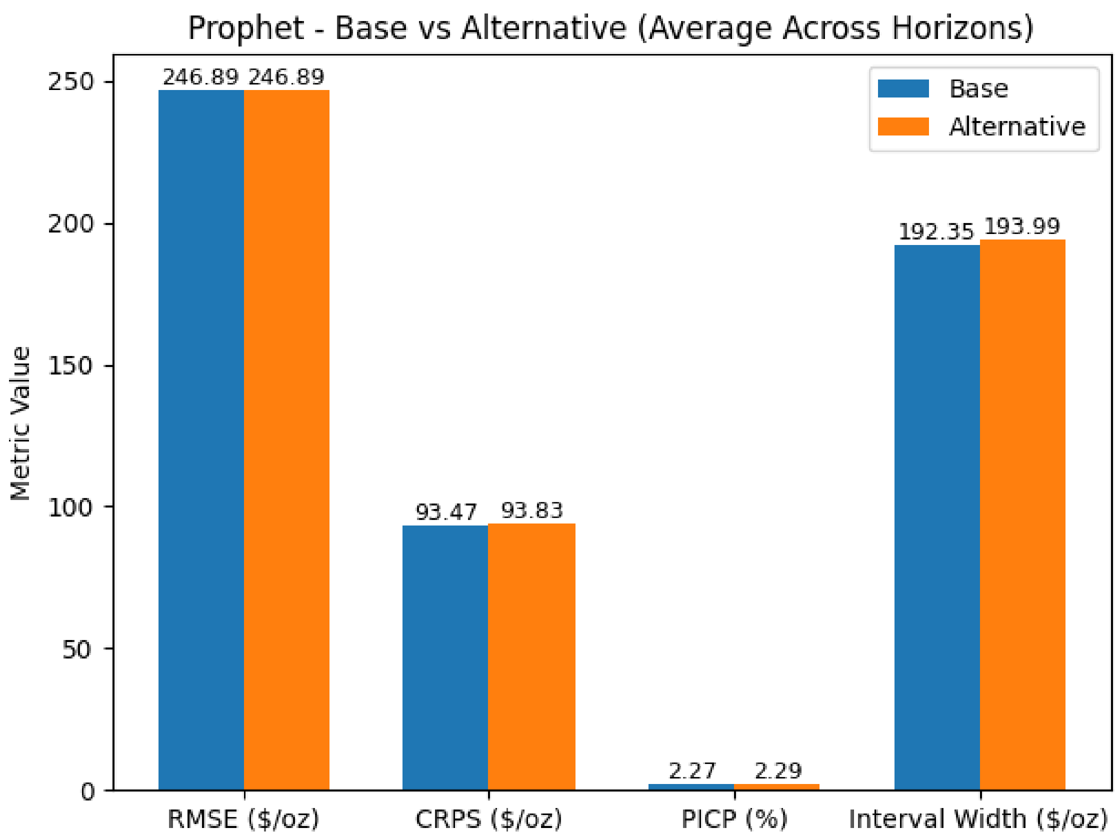

Figure 5. Bar graph of Prophet model metric values

6.2 Quantile Regression Forest Model

The next model trained was the QRF model, a tree-based model previously mentioned in this thesis. This model was created with the following configurations:

- n_estimators=300,
- min_samples_leaf=5
- random_seed=42

The n_estimators parameter refers to the number of decision trees the model will build. In this case, 300 decision trees were built, this number was chosen as a suitable balance between forecasting accuracy and computational resource usage. Typically, increasing the number of trees leads to higher accuracy and stability, however it does also increase computation time and load. The samples_leaf parameter refers to the number of samples a leaf can split into. In this case, this means that each decision path must have at least 5 samples. This was chosen to prevent the model from creating overly specific rules for a small subset of the dataset, which would otherwise lead to overfitting. The last parameter refers to the random seed for reproducibility. Having a fixed seed means that this experiment would always be reproduceable and have consistent results which is essential for experimentation and comparisons. The value 42 is common convention and standard practice.

Figure 6 below shows a comparison of the model's probabilistic forecast of the value of gold for the last 10 trading days of the datasets. From the charts, we can observe that the model had an interval width of around $70 using the base dataset for the subset of those last 10 days, whereas using the alternative dataset resulted in a slightly wider interval width of approximately $90. This increased interval width is likely the driving force behind the increased coverage in the alternative dataset, meaning more of the actual values fall within the forecasted range. Interestingly, it is very clear that there appears to be a clear directional mismatch between the forecasted values and actual values, specifically around the 2nd trading day horizon where the forecasted values drop, while the actual values increase. This phenomenon is later corrected in later daily horizons with the alternative dataset model producing a sharp upward correction around the 6th trading day horizon.

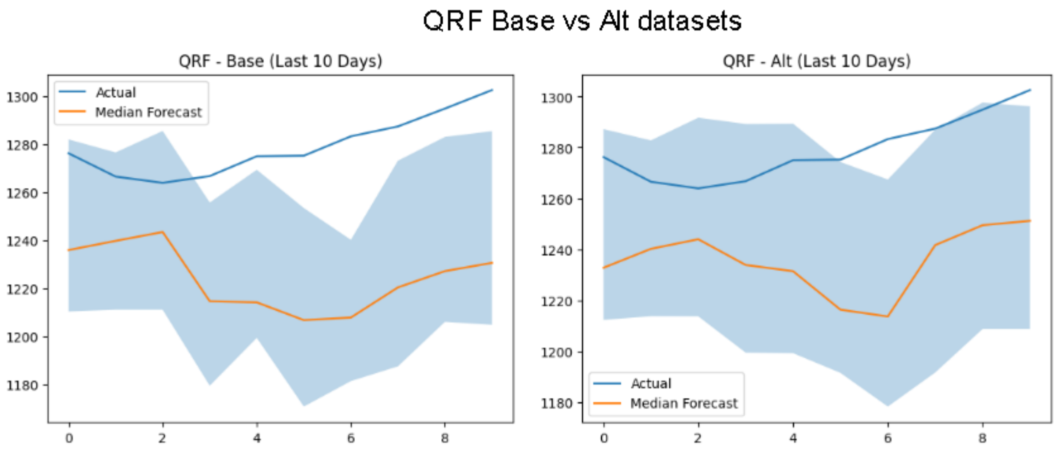

Figure 6. QRF forecast range of t+10 horizon, base vs. alternative dataset, last 10 trading days

For more detailed information on the performance of the QRF model, we dive deeper into the actual metric values across the entire datasets. Figure 7 below shows two tables comparing the model's values for each metric, for each daily horizon, between the base and alternative datasets.

QRF - Base Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	2.959022	0.767705	26.258569	11.893238
2	4.604141	0.739377	39.828304	18.577658
3	5.678306	0.688385	47.378051	21.873345
4	6.647257	0.705382	55.111560	25.871401
5	7.613785	0.711048	62.674106	29.570219
6	8.899928	0.677054	72.626090	34.883935
7	10.146901	0.628895	76.664844	38.367284
8	11.396838	0.628895	80.105504	42.327162
9	12.856602	0.589235	85.687804	46.836821
10	13.871501	0.583569	88.240945	50.003514
Average	8.467428	0.671955	63.457578	32.020458

QRF - Alternative Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	2.929655	0.781870	27.465148	11.771439
2	4.379798	0.756374	40.758674	17.585171
3	5.395969	0.722380	49.188372	20.836140
4	6.695234	0.739377	61.085451	26.104200
5	7.919101	0.728045	66.352480	31.355538
6	9.365589	0.651558	71.284383	36.954663
7	10.812527	0.611898	75.419522	41.461386
8	12.143329	0.592068	80.171462	45.888265
9	13.593150	0.541076	86.804093	51.171832
10	13.717802	0.572238	92.491575	52.217755
Average	8.695215	0.669688	65.102116	33.534639

Figure 7. QRF model avg evaluation metrics per daily horizon of the test data subset (last 20% of 2007-2017 dataset). Base vs Alternative datasets

We can observe from the above figure that on average, the PICP, measuring coverage, has a value of around 0.6719 when using the base dataset while the alternative dataset resulted in a value of 0.6696. This means that on average, actual values fell within the forecast range around 67% of the time, a significant improvement from the results of the baseline prophet model. This is also a much closer result to the 80% prediction interval used in the research design. Consequently, the RMSE shows a lower value of around 32 and 33.5 points for the base and alternative datasets respectively, while the CRPS also has a low score of approximately 8.47 points in the base dataset and 8.70 in the alternative dataset. It is also worth noting that the RMSE progressively increases across both datasets as the daily horizon increases. This is expected behaviour since uncertainty and in conjunction, the margin of error compounds overtime in financial forecasting. All of these metrics can be visualized in a bar graph for comparison in figure 8 below.

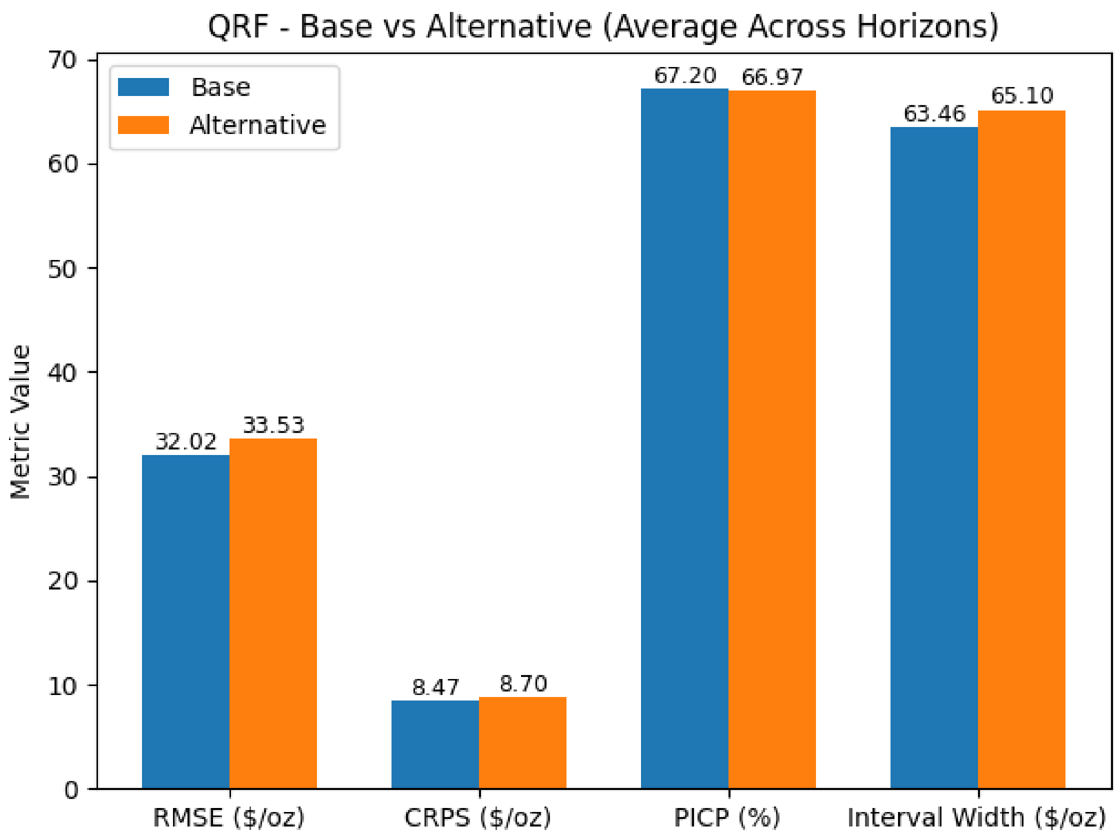

Figure 8. Bar graph of QRF model metric values

6.3 XGBoost Model

The final model trained and tested was the XGBoost model. This model was designed with the following configurations:

- objective="reg:quantileerror"
- quantile_alpha= [0.1, 0.5, 0.9]
- n_estimators=300,
- learning_rate=0.05,
- max_depth=4,
- random_state=42

While there aren't optimal values for these configurations, it is important to keep the bigger picture and the objective of the research in mind. These values were chosen as they constitute the right balance between complexity, computational feasibility, and a balanced bias-variance tradeoff.

As previously mentioned, the XGBoost model is another tree-based model originally optimized for point value predictions, however in this case, its objective function is changed to quantile loss or quantile error to predict quantile intervals. This tells XGBoost to optimize for quantile regression in order to estimate conditional quantiles of the target distribution. For simplicity, “conditional quantiles” means producing quantiles or intervals on the condition of the provided features.

Quantile alpha simply refers to the quantiles that will be estimated, in this case the 10^{th}, 50^{th} (mean), and 90^{th} percentiles, leading to an 80% prediction interval as was stated in the methodologies section.

“N estimators” refers to the number of trees built similar to the QRF model, in this case that value was set to 300 in order to balance between forecasting accuracy and resource overhead as well as to make the models comparable to each other.

The learning rate here is essentially the impact of each tree’s prediction which means how much each new tree contributes to the overall prediction. The higher this number, the bigger the impact each tree has on the overall prediction, this means the model learns faster but risks overfitting. Conversely, a lower learning rate means the model learns slower but more carefully, resulting in a more thoughtful output. This number was chosen to ensure gradual learning and to decrease the risk of overfitting. Furthermore, a lower learning rate requires a higher number of trees to obtain a good performance, which was the strategy adopted in this case.

Max depth limits how deep each tree can grow. Lower max depth leads to high bias or underfitting while higher max depth leads to high variance or overfitting. A delicate balance between the two is important to prevent either extreme fitting of data, which is why a value of 4 was chosen. Finally, random state refers to the same concept discussed in QRF for reproducibility.

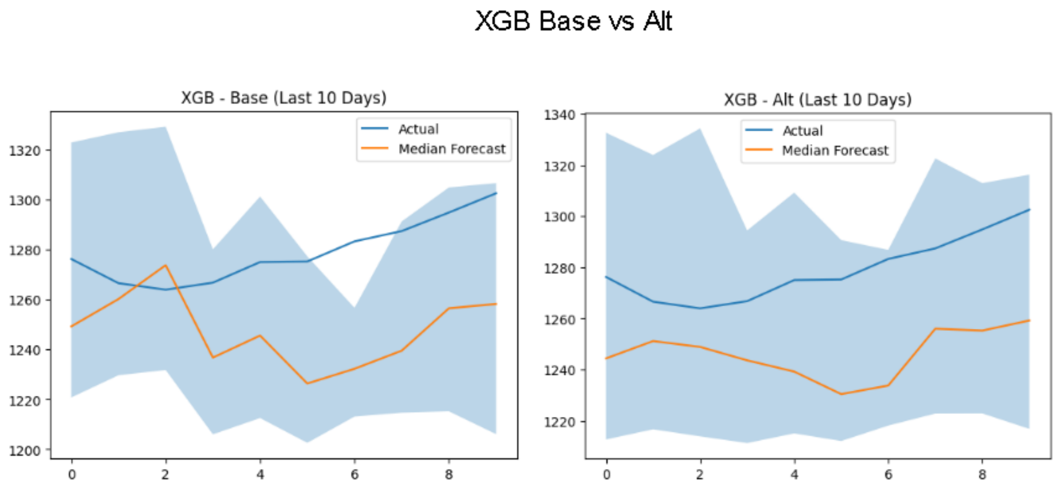

Figure 9. XGBoost forecast range of t+10 horizon, base vs. alternative dataset, last 10 trading days.

The figure above showcases a comparison between the predicted intervals of the XGBoost model while using the base dataset vs the alternative dataset for the last 10 trading days. Immediately the difference in interval width is very apparent, with the base dataset starting out with an interval width of around $100, while the alternative dataset consistently has a wider interval width of around $120, signalling more uncertainty. As a result of the wider interval width, one can observe that the alternative dataset model has a much better coverage of actual values falling within prediction ranges compared to the base dataset. While normally a wider interval width results in a less useful forecast, even if the coverage is better, in this case, the forecast still shows promise despite the increased interval width. In order to get a clearer understanding of the model's performance, we must first analyse each metrics average results across the entire dataset. Figure 10 below shows a tabular comparison between the metric values of the XGBoost model when trained on the base dataset vs the alternative dataset, across each of the 10 daily horizons.

XGB - Base Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	4.723097	0.869688	48.813702	21.559425
2	5.798786	0.662890	36.120716	28.255293
3	7.058120	0.750708	49.955948	32.498093
4	7.900584	0.722380	55.693497	38.174856
5	7.578887	0.728045	63.445175	31.089710
6	8.453342	0.679887	64.608582	36.050862
7	9.241134	0.696884	76.767349	39.803008
8	9.804961	0.677054	83.088600	40.171865
9	10.698711	0.708215	93.155434	45.217135
10	11.636238	0.668555	88.194298	47.320442
Average	8.289386	0.716431	65.984329	36.014069

XGB - Alternative Dataset

Horizon	CRPS	PICP	Interval_Width	RMSE
1	4.490369	0.835694	38.756893	21.434129
2	5.907307	0.739377	45.231651	27.366016
3	7.500905	0.696884	45.557587	34.729558
4	8.720028	0.767705	60.185692	46.491174
5	7.567679	0.756374	65.344910	31.423321
6	8.433063	0.764873	77.217590	35.136013
7	9.402922	0.725212	75.561783	41.766766
8	10.530546	0.736544	100.833595	41.468412
9	10.417067	0.739377	92.241615	43.621314
10	11.396741	0.804533	102.716347	49.666977
Average	8.436663	0.756657	70.364769	37.310368

Figure 10. XGBoost model avg evaluation metrics per daily horizon of the test data subset (last 20% of 2007-2017 dataset). Base vs Alternative datasets

We can observe from the above figure that on average, the PICP, measuring coverage, has a value of around 0.7164 when using the base dataset while the alternative dataset resulted in a slightly higher value of 0.7566. This means that on average, actual values fell within the forecast range around 72% of the time while using the base dataset vs 76% of the time while using the alternative dataset, a slight improvement from the reported 67% results of the QRF model. This is the highest result and the closest to the 80% prediction interval used in the research design, out of all models trained. The RMSE shows a value of around 36 and 37.3 $/oz for the base and alternative datasets respectively, while the CRPS also has a low score of approximately 8.29 points in the base dataset and 8.43 in the alternative dataset. It is also worth noting that similarly to the QRF model results, the RMSE of the XGBoost models progressively increases across both datasets as the daily horizon increases. All of these metrics can be visualized in a bar graph for comparison in figure 11 below.

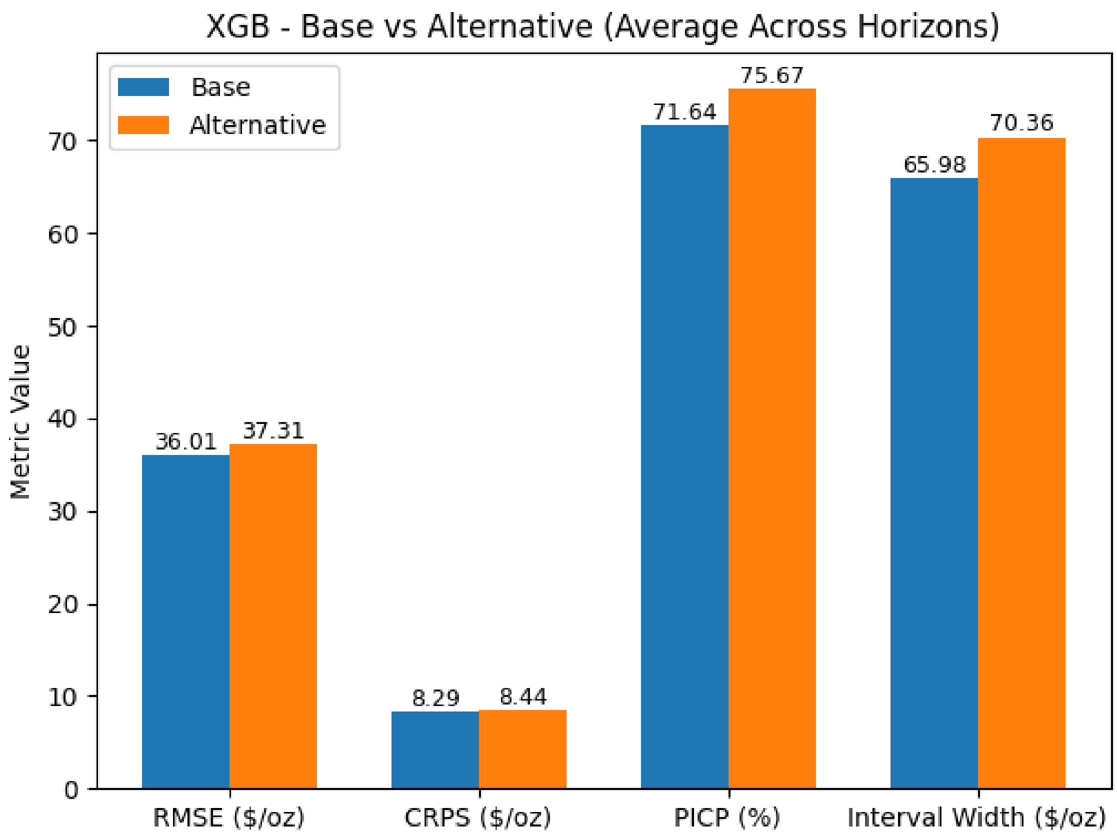

Figure 11. Bar graph of XGBoost model metric values

While other models showed relatively similar results between the metric values produced by using the base dataset vs the alternative dataset, the results of the XGBoost model evaluations tell a slightly different story with the interval width in particular being noticeably higher with the alternative dataset.

A further detailed interpretation and comparative analysis will be discussed in the next section.

7 DISCUSSION

The results outlined in the previous section show a clear hierarchical ranking of performance between the three models employed in this research. It was evident that the tree-based machine learning models, Quantile Regression Forest and XGBoost, significantly outperformed the linear Prophet model across all evaluation metrics. Prophet produced significantly higher RMSE scores and very poor prediction interval coverage, indicating that it struggled to accurately model the dynamics of gold price movements across each of the 10 evaluation daily horizons. In contract, the tree-based machine learning models produced not only more accurate point forecasts and more reliable probabilistic predictions but also a tighter interval width and far better coverage as is summarized in Figure 12 below.

Dataset	Model	CRPS	PICP	Interval_Width	RMSE
Base	QRF	8.467428	0.671955	63.457578	32.020458
Base	XGB	8.289386	0.716431	65.984329	36.014069
Base	Prophet	92.894312	0.024646	192.071772	244.675406
Alt	QRF	8.695215	0.669688	65.102116	33.534639
Alt	XGB	8.436663	0.756657	70.364769	37.310368
Alt	Prophet	92.351029	0.025212	192.594995	244.675406

Figure 12. Comparison of model performance across metrics

The higher performance of tree-based ensemble methods come as no surprise as it is well documented in existing financial forecasting literature such as Gu et al. (2020). Random forest and gradient boosting models are well known for their ability to capture nonlinear relationships and complex interactions between variables. It is worth re-iterating once more than predicting financial markets is an extremely difficult endeavour as they are influenced by a wide range of factors that affect each other such as macroeconomic indicators, market volatility, and investor sentiment.

Since Prophet relies on additive trends and seasonalities, it is an extremely effective tool in business forecasting contexts such as predicting retail sales or web traffic. However, it fails to capture the complexities of financial markets and their stochastic, or random, behaviour. Gold prices are characterized by extreme sensitivity to macroeconomic factors which lead to sudden volatility, but because Prophet assumes a relatively smooth underlying trend structure, it is less capable of adapting to abrupt changes in market conditions. This likely explains the significantly higher forecasting errors observed for the Prophet model.

If we look at previously reported evaluation metrics from other experiments, reported RMSE values in the literature vary significantly depending on dataset and methodology. For example, studies such as Cohen, G. and Aiche, A. (2023). and Alharbi, M.H. (2024) report lower RMSE values for tree-based models compared to baseline approaches. This is consistent with our own findings in this research, where the tree-based models (QRF and XGBoost) also achieved lower RMSE values, 32 and 36 $/oz respectively, compared to the Prophet model which had an average value of approximately 248 $/oz. Although a direct comparison is not realistic due to differences in data preprocessing, forecasting horizons, and evaluation frameworks.

If we switch focus to comparing the two tree-based machine learning models, we can observe that there is a relatively small difference in performance between them. Both models resulted in very similar values across the evaluated metrics, with only slight differences depending on the forecasting horizon. The Quantile Regression Forest typically resulted in lower RMSE and average interval width values, suggesting it is slightly better at forecasting point values with higher confidence. XGBoost on the other hand often produced higher PICP or coverage scores closer to the expected prediction interval and marginally better CRPS scores, indicating it is more suited to probabilistic forecasting with decent accuracy. If we consider CRPS as the primary evaluation metric as was mentioned in the methodologies section, then we can surmise that out of the three models, on average, XGBoost performs the best in this specific scenario. In general, both tree-based models are well suited to the task of gold price forecasting. The choice between them may depend on practical considerations such as computational efficiency, interpretability, or ease of implementation.

This brings us to the other comparative aspect of this research, whether alternative datasets improve forecasting accuracy or not. We will dissect the results for each Model. The results in Figure 10 show that for the QRF, using alternative dataset has actually resulted in worse scores across all metrics. The RMSE score was 33.5 using alternative dataset compared to 32 in the base dataset. The interval width was higher at 65.1 signalling less confidence using the alternative dataset compared to 63.4 using the base dataset. The CRPS had a result of 8.69 using the alternative dataset compared to 8.46 with the base dataset, a decline of approximately 2.6% in performance. Similarly, the results for the XGB model were slightly better using the base dataset than the alternative dataset, with the RMSE scoring 36 vs 37.3 and the CRPS scoring 8.28 vs 8.43 respectively. Interestingly, XGB had a noticeably better PICP or coverage score of 0.756 using the alternative dataset than the base dataset at 0.716. However, this is primarily due to the increased interval width from 65 to 70, signalling a wider net and decreased forecasting confidence. Overall, XGB's forecasting performance declined by around 1.75% when using alternative dataset compared with the evaluation results of the base dataset. The results of the prophet model were not relevant in this section since the model does not take features into account. Any slight differences in results between alternative and base dataset usage is purely stochastic and does not carry any scientific relevance.

It is worth noting that for all metrics and models, the difference is very minimal and hardly impactful, however the author of this research felt the need to still acknowledge it given the emphasis placed on incremental improvements earlier in the research paper.

There are several possible explanations to these outcomes. First, there may be a high level of noise in alternative data variables compared to the predictive value they bring. Search trends and news sentiment may capture general public interest in gold markets, but this information may not translate directly into short-term price movements as price action is more highly influenced by macroeconomic indicators, trading activity, and geopolitical events.

Second, the time horizon used in this study may influence the usefulness of alternative data. Sentiment and search activity may have predictive value over longer time horizons causing gradual changes in investor behaviour or macroeconomic expectations. However, within a ten day forecasting window, these signals may not have enough weight to meaningfully affect model predictions.

Third, the transformation and aggregation of the sentiment data may have reduced the informational content of the original dataset. In order to align the data temporally and avoid data leakage, the sentiment data were aggregated to daily values and lagged before being integrated into the model features. While this step is necessary for methodological correctness, it may also reduce the granularity and value of the information available to the models.

At this stage, it is worth mentioning some limitations that were encountered while conducting this research. Firstly, it was not feasible to conduct the research in a true walk-forward approach due to computational resource limitations, walk-forward would have meant re-training each of the three models for every time step around the 10 year datasets, two times for each dataset (base and alternative). This would have resulted in several thousand model re-trainings which exceed the computational capacity available. Instead, a simple chronological split was used while ensuring no data leakage. This setup was significantly simpler and is a common approach to forecasting among predictive analytics researchers. Another limitation that was unavoidable was the simplification of the news sentiment dataset by aggregating and reducing the data to a simple net sentiment score and headline count. This was a necessary step in order to be able to align the datasets on dates. However, in doing so, one could argue that sentiment data loses a lot of its context and intensity or impact. Based on the above reported limitations and constraints, it is the opinion of this author that future research should endeavour to 1. Implement true walk-forward validation to mimic real world performance 2. Decrease the forecasting horizon to focus on hours instead of days and 3. Increase the granularity of the alternative datasets in order to match it to price data on an hourly or 10 minute basis.

Finally, to dive deeper into the meaning behind the results it was necessary to compare the importance of the features used. Starting off the feature analysis with the QRF model features, figure 13 below shows that the two primarily dominant features are the gold price 10 day moving average (Gold_MA_10) and the daily gold price closing (Gold_Close), accounting for approximately 72% of the total feature importance. Signalling that traditional financial metrics are still the biggest drivers of financial asset price forecasting, considering that all the features from alternative data such as the VIX, DXY, Google Trends, and news sentiment score significantly lower on the feature importance scale. In simple terms, this means that alternative data metrics have little to no impact on the predictive model's decision making when it comes to forecasting gold price movements over a 10 day horizon.

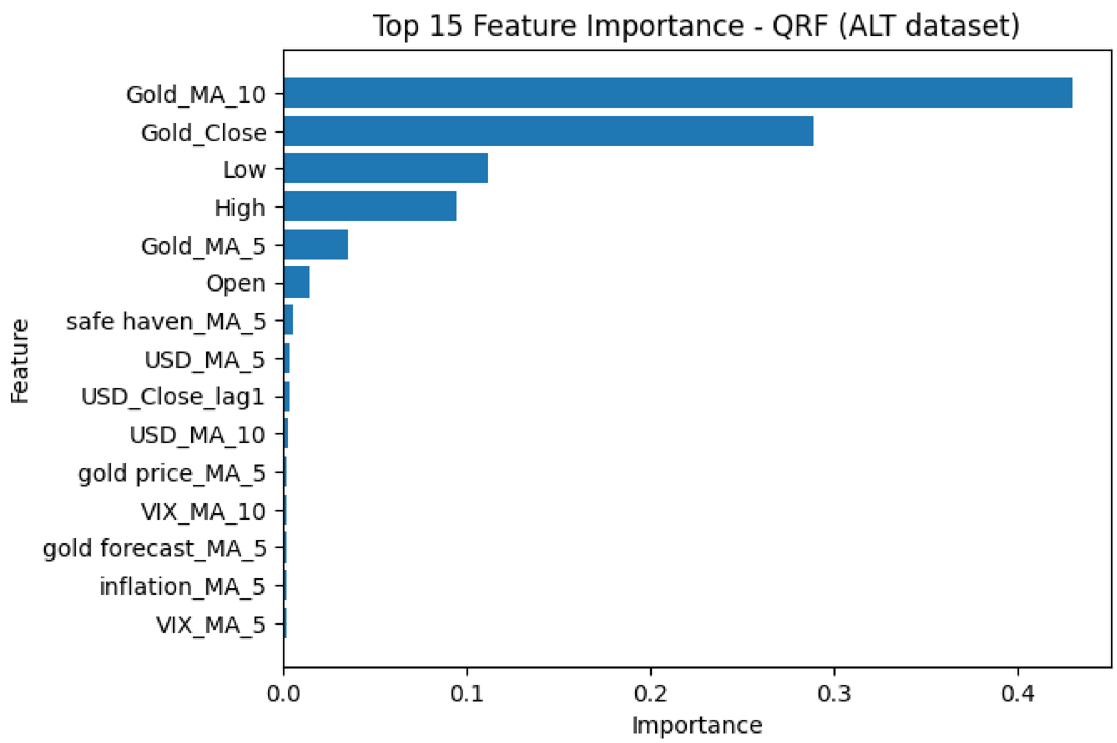

Figure 13. QRF model features importance

Similar results we discovered in the XGB model's feature analysis as is show in Figure 14 below. The two primarily dominant features were the daily price high and low, jointly making up around 61% of the total feature importance across all features. This aligns with expectations given the model's probabilistic forecasting superiority. Another interesting find is that the next two most important features are the gold price 10 day moving average (Gold_MA_10) and the daily gold price closing (Gold_Close), which were the top two features in the QRF model. While the daily gold price low and high, which are the top two features in the XGB model, were the 3rd and 4th most important features in the QRF model. Between both models, these 4 features make up around 92% of the total feature importance in the QRF model and around 78% of the total feature importance in the XGB model. An overwhelming majority in both cases, further emphasising the lack of impact of the alternative dataset features.

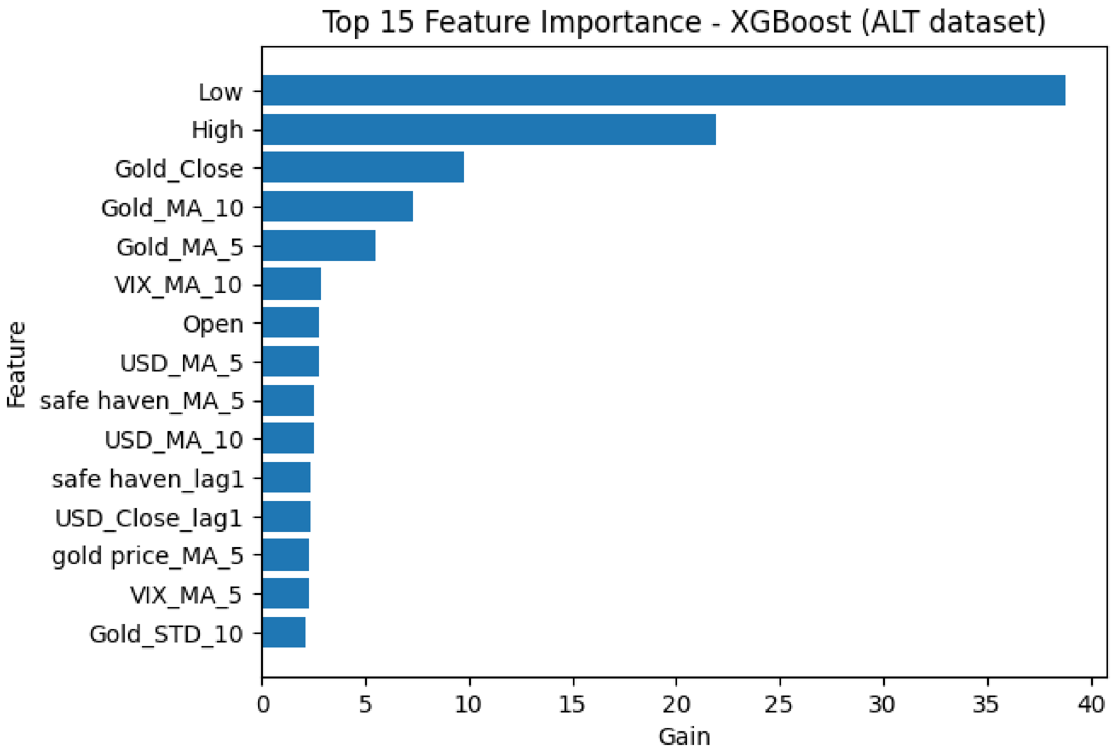

Figure 14. XGBoost model feature importance

Despite the lack of improvement from using alternative data, these results remain extremely valuable. Negative findings are important in scientific research, particularly in the context of financial forecasting where accurate prediction methodologies and tools that provide consistent results continue to be elusive. The results suggest that not all forms of alternative data are equally useful for predicting gold prices, and that careful feature engineering and data selection remain critical.

8 CONCLUSION

In conclusion, this research has attempted to explore the impact of utilizing alternative datasets along with machine learning models for the purpose of forecasting the price of one of the world's oldest stores of value, gold. The aim of this endeavour was to explore whether alternative data that has been processed and analysed by machine learning models, can enhance the prediction of short term (2 - 4 weeks) gold price movements. The primary focus was on the predictive value of alternative data, not on building a deployable trading system. Three types of machine learning models were employed: Prophet, a linear time series forecasting model, and two tree-based models, QRF and XGBoost.

The results of the analysis have shown that on average there was an approximate 2% decline in predictive performance based on average CRPS scores, when using alternative datasets versus relying solely on traditional financial metrics. This was largely due to the increased noise introduced by those metrics as well as the length of the forecasting horizon, spanning over 10 trading days. Since the prediction performance decreased, one can directly address the research question of meaningful impact by stating that alternative data did not provide a statistically meaningful improvement to predictive performance.

These results have answered the remaining research questions of this thesis by showing that, in this case, alternative data has not improved the accuracy of short-term gold price forecasts when compared to using price-based indicators alone. Out of the three models used, and if we consider CRPS as the primary evaluation metric, it would be clear that XGBoost performs best when incorporating alternative data for predicting gold price movements.

As for the importance of alternative dataset features, results have revealed that alternative data, such as Google Trends data or sentiment analysis from financial news, have limited importance and a weak relationship with gold price movements as the vast majority of feature importance was driven by traditional financial metrics such as moving average, daily price lows, highs, and closing.

9 AI DECLARATION

Generative AI was used to brainstorm ideas for topics for the purpose of writing this paper. Specifically, the AI model “GPT-5.1”. Generative AI was also used to brainstorm ideas on the content, flow, and sentence structure in this paper. It was consulted on the feasibility of the proposed thesis and its relevance.

All content in the final version has been written and evaluated by the author to ensure accuracy, clarity, and academic integrity.

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APPENDICES

Appendix 1

```
from pytrends.request import TrendReq
import pandas as pd
import time
import random
from datetime import datetime, timedelta
import os


start_time = datetime.now()
print(f"Starting at {start_time}")

# ----------------------------
# CONFIG
# ----------------------------

keywords = [
    'gold price',
    'gold forecast',
    'buy gold',
    'inflation',
    'safe haven'
]

start_date = datetime(2008, 1, 1)
end_date = datetime(2017, 12, 31)
window_size = 180  # ~6 months

OUTPUT_FILE = 'google_trends_gold_keywords_daily_2008_2017.csv'

BASE_SLEEP = 10
RATE_LIMIT_SLEEP_MIN = 120
RATE_LIMIT_SLEEP_MAX = 300

# ----------------------------
# INIT PYTRENDS
# ----------------------------

pytrend = TrendReq(
    hl='en-US',
    tz=360,
    timeout=(10, 25)
)

# ----------------------------
# LOAD EXISTING DATA (if any)
# ----------------------------

if os.path.exists(OUTPUT_FILE):
    final_df = pd.read_csv(OUTPUT_FILE, parse_dates=['date'])
    completed_keywords = set(final_df.columns) - {'date'}
    print(f"Resuming. Found existing keywords: {completed_keywords}")
else:
```

```
    final_df = None
    completed_keywords = set()
    print("Starting fresh run.")

# ----------------------------
# MAIN LOOP
# ----------------------------

for kw in keywords:

    if kw in completed_keywords:
        print(f"Skipping already completed keyword: {kw}")
        continue

    print(f"\nProcessing keyword: {kw}")
    all_dfs = []
    current_start = start_date

    while current_start < end_date:
        current_end = min(current_start + timedelta(days=window_size), end_date)
        timeframe = f"{current_start.strftime('%Y-%m-%d')} {current_end.strftime('%Y-%m-%d')}"

        print(f"  Fetching {timeframe}")

        try:
            pytrend.build_payload(
                kw_list=[kw],
                timeframe=timeframe,
                geo=''
            )

            df = pytrend.interest_over_time()

            if not df.empty:
                df = df[[kw]]
                all_dfs.append(df)

            time.sleep(BASE_SLEEP)

        except Exception as e:
            if "429" in str(e):
                wait = random.randint(RATE_LIMIT_SLEEP_MIN, RATE_LIMIT_SLEEP_MAX)
                print(f"⛔ Rate limited. Sleeping for {wait} seconds...")
                time.sleep(wait)
                continue
            else:
                print(f"⚠ Error: {e}. Retrying in 10s...")
                time.sleep(10)
                continue

        current_start = current_end + timedelta(days=1)

    # ----------------------------
    # MERGE & SAVE CHECKPOINT
    # ----------------------------

    if not all_dfs:
        print(f"No data collected for keyword: {kw}")
        continue
```

```
    kw_df = pd.concat(all_dfs)
    kw_df = kw_df[~kw_df.index.duplicated(keep='first')]
    kw_df.reset_index(inplace=True)

    if final_df is None:
        final_df = kw_df
    else:
        final_df = final_df.merge(kw_df, on='date', how='outer')

    final_df.sort_values('date', inplace=True)

    # SAVE AFTER EACH KEYWORD
    final_df.to_csv(OUTPUT_FILE, index=False)
    print(f"✅ Saved progress after keyword: {kw}")

# ----------------------------
# DONE
# ----------------------------

end_time = datetime.now()
print(f"Ending at {end_time}")

elapsed = end_time - start_time


print("\nAll keywords processed.")
print(f"Final dataset saved to: {OUTPUT_FILE}")
print(f"Script took (hh:mm:ss) {elapsed} to run")
print(final_df.head())
Starting at 2026-01-29 13:54:26.967318
Starting fresh run.

Processing keyword: gold price
 Fetching 2008-01-01 2008-06-29
 Fetching 2008-06-30 2008-12-27
 Fetching 2008-12-28 2009-06-26
 Fetching 2009-06-27 2009-12-24
 Fetching 2009-12-25 2010-06-23
 Fetching 2010-06-24 2010-12-21
 Fetching 2010-12-22 2011-06-20
 Fetching 2011-06-21 2011-12-18
 Fetching 2011-12-19 2012-06-16
 Fetching 2012-06-17 2012-12-14
 Fetching 2012-12-15 2013-06-13
 Fetching 2013-06-14 2013-12-11
 Fetching 2013-12-12 2014-06-10
 Fetching 2014-06-11 2014-12-08
 Fetching 2014-12-09 2015-06-07
 Fetching 2015-06-08 2015-12-05
 Fetching 2015-12-06 2016-06-03
 Fetching 2016-06-04 2016-12-01
 Fetching 2016-12-02 2017-05-31
 Fetching 2017-06-01 2017-11-28
 Fetching 2017-11-29 2017-12-31
✅ Saved progress after keyword: gold price

Processing keyword: gold forecast
 Fetching 2008-01-01 2008-06-29
```

```
  Fetching 2008-06-30 2008-12-27
  Fetching 2008-12-28 2009-06-26
  Fetching 2009-06-27 2009-12-24
  Fetching 2009-12-25 2010-06-23
  Fetching 2010-06-24 2010-12-21
  Fetching 2010-12-22 2011-06-20
  Fetching 2011-06-21 2011-12-18
  Fetching 2011-12-19 2012-06-16
  Fetching 2012-06-17 2012-12-14
  Fetching 2012-12-15 2013-06-13
  Fetching 2013-06-14 2013-12-11
  Fetching 2013-12-12 2014-06-10
  Fetching 2014-06-11 2014-12-08
  Fetching 2014-12-09 2015-06-07
  Fetching 2015-06-08 2015-12-05
  Fetching 2015-12-06 2016-06-03
  Fetching 2016-06-04 2016-12-01
⛔ Rate limited. Sleeping for 197 seconds...
  Fetching 2016-06-04 2016-12-01
  Fetching 2016-12-02 2017-05-31
  Fetching 2017-06-01 2017-11-28
  Fetching 2017-11-29 2017-12-31
✅ Saved progress after keyword: gold forecast

Processing keyword: buy gold
  Fetching 2008-01-01 2008-06-29
  Fetching 2008-06-30 2008-12-27
  Fetching 2008-12-28 2009-06-26
  Fetching 2009-06-27 2009-12-24
  Fetching 2009-12-25 2010-06-23
  Fetching 2010-06-24 2010-12-21
  Fetching 2010-12-22 2011-06-20
  Fetching 2011-06-21 2011-12-18
  Fetching 2011-12-19 2012-06-16
  Fetching 2012-06-17 2012-12-14
  Fetching 2012-12-15 2013-06-13
  Fetching 2013-06-14 2013-12-11
  Fetching 2013-12-12 2014-06-10
  Fetching 2014-06-11 2014-12-08
  Fetching 2014-12-09 2015-06-07
  Fetching 2015-06-08 2015-12-05
  Fetching 2015-12-06 2016-06-03
  Fetching 2016-06-04 2016-12-01
  Fetching 2016-12-02 2017-05-31
  Fetching 2017-06-01 2017-11-28
  Fetching 2017-11-29 2017-12-31
✅ Saved progress after keyword: buy gold

Processing keyword: inflation
  Fetching 2008-01-01 2008-06-29
  Fetching 2008-06-30 2008-12-27
  Fetching 2008-12-28 2009-06-26
  Fetching 2009-06-27 2009-12-24
  Fetching 2009-12-25 2010-06-23
  Fetching 2010-06-24 2010-12-21
  Fetching 2010-12-22 2011-06-20
  Fetching 2011-06-21 2011-12-18
  Fetching 2011-12-19 2012-06-16
  Fetching 2012-06-17 2012-12-14
```

```
 Fetching 2012-12-15 2013-06-13
 Fetching 2013-06-14 2013-12-11
 Fetching 2013-12-12 2014-06-10
 Fetching 2014-06-11 2014-12-08
 Fetching 2014-12-09 2015-06-07
 Fetching 2015-06-08 2015-12-05
 Fetching 2015-12-06 2016-06-03
 Fetching 2016-06-04 2016-12-01
 Fetching 2016-12-02 2017-05-31
 Fetching 2017-06-01 2017-11-28
 Fetching 2017-11-29 2017-12-31
✅ Saved progress after keyword: inflation

Processing keyword: safe haven
 Fetching 2008-01-01 2008-06-29
 Fetching 2008-06-30 2008-12-27
 Fetching 2008-12-28 2009-06-26
 Fetching 2009-06-27 2009-12-24
 Fetching 2009-12-25 2010-06-23
 Fetching 2010-06-24 2010-12-21
 Fetching 2010-12-22 2011-06-20
 Fetching 2011-06-21 2011-12-18
 Fetching 2011-12-19 2012-06-16
 Fetching 2012-06-17 2012-12-14
 Fetching 2012-12-15 2013-06-13
 Fetching 2013-06-14 2013-12-11
 Fetching 2013-12-12 2014-06-10
 Fetching 2014-06-11 2014-12-08
 Fetching 2014-12-09 2015-06-07
 Fetching 2015-06-08 2015-12-05
 Fetching 2015-12-06 2016-06-03
 Fetching 2016-06-04 2016-12-01
 Fetching 2016-12-02 2017-05-31
 Fetching 2017-06-01 2017-11-28
 Fetching 2017-11-29 2017-12-31
✅ Saved progress after keyword: safe haven
Ending at 2026-01-29 14:16:55.767603

All keywords processed.
Final dataset saved to: google_trends_gold_keywords_daily_2008_2017.csv
Script took (hh:mm:ss) 0:22:28.800285 to run
        date  gold price  gold forecast  buy gold  inflation  safe haven
0 2008-01-01          27            100        58         41           0
1 2008-01-02          45             49        44         55           0
2 2008-01-03          54             76        43         55           0
3 2008-01-04          45             83        49         60          45
4 2008-01-05          34             47        62         50           0
```

```python
import pandas as pd
from datetime import datetime
import re

# Load dataset
df = pd.read_csv("gold-dataset-sinha-khandait.csv")

# Regex pattern to find YYYY-MM-DD in URLs
date_pattern = re.compile(r"\d{4}-\d{2}-\d{2}")

def fix_date(row):
  original_date = str(row["Dates"])
  url = str(row["URL"])

  # Regex pattern for YYYY-MM-DD
  valid_pattern = re.compile(r"\d{4}-\d{2}-\d{2}")
  url_pattern = re.compile(r"\d{4}-\d{2}-\d{2}")

  # Try to extract a valid date from the URL
  match = url_pattern.search(url)
  extracted_date = match.group(0) if match else None

  def normalize(date_str):
    try:
      return pd.to_datetime(date_str, errors="coerce").strftime("%Y-%m-%d")
    except Exception:
      return None

  normalized = normalize(original_date)

  # Check if normalized year is reasonable
  if normalized:
    year = int(normalized[:4])
    if 1900 <= year <= datetime.now().year:
      # Valid year, keep it
      return pd.Series([normalized, False])
    else:
      # Year out of range, replace with URL date if available
      if extracted_date:
        return pd.Series([extracted_date, True])
      else:
        return pd.Series([normalized, False])
  else:
    # If normalization failed, try URL
    if extracted_date:
      return pd.Series([extracted_date, True])
    else:
      return pd.Series([original_date, False])

# Apply function to each row
df[["new_date", "date_modified"]] = df.apply(fix_date, axis=1)
df.drop(columns=['Dates'], inplace=True)
df.rename(columns={"new_date": "date"}, inplace=True)
# Save cleaned dataset
df.to_csv("gold-sentiment-dataset-cleaned.csv", index=False)
```

```
print("☑ Cleaning complete. Saved as gold-dataset-cleaned.csv")
print(df[["date", "date_modified"]].head())
```

```
C:\Users\AbdoZ\AppData\Local\Temp\ipykernel_27520\1942584045.py:25: UserWarning: Parsi
ng dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True
` or specify a format to silence this warning.
  return pd.to_datetime(date_str, errors="coerce").strftime("%Y-%m-%d")
✅ Cleaning complete. Saved as gold-dataset-cleaned.csv
        date  date_modified
0  2016-01-28          False
1  2017-09-13          False
2  2016-07-26          False
3  2018-02-28          False
4  2017-06-09          False
```